Nithin Manoj R D

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.91
Velammal Bodhi Campus, Ponneri	Class XII (CBSE)	2023	93.60%
Aristo Public School, Cuddalore	Class X (CBSE)	2021	94.40%

EXPERIENCE

AI Technical Intern -- Wizlearnr

- Developed an **ASR-in-the-loop Telugu TTS validation pipeline**, improving clean-audio generation from **61% 97%**; built phoneme-aware pronunciation assessment using **IPA + Weighted Phonetic Edit Distance** with a custom Telugu phoneme-weight map.
- Optimized **IndicConformer Telugu ASR** for edge deployment using mixed-precision post-training quantization, reducing model size **82.7% (522 90 MB)** with negligible recognition-accuracy degradation.
- Optimized **Sarvam-30B MoE** inference using GPTQ/vLLM; built a reproducible latency-throughput-energy benchmark and traced an unexpected quantization slowdown to **vLLM eager-mode execution**, restoring the optimized runtime path.
- Proposed a **UMLS concept-centric RAG** architecture for women's-health QA, mapping patient queries to clinical concepts with QuickUMLS and retrieving structured disease representations for grounded generation with **MedGemma-27B**.

SKILLS

- **Programming:** C, C++, Python (PyTorch, NumPy, pandas, scikit-learn, Matplotlib), R, SQL
- **Scientific Computing & Shell:** Bash (grep, sed, awk), Jupyter
- **ML/AI:** Machine Learning, Deep Learning, LLM Inference, Model Quantization, RAG
- **DevOps & Tools:** Git, GitHub, Docker, CI/CD, Hugging Face
- **Cloud:** Google Cloud Platform (GCP)

RELEVANT COURSES

- **Math Domain** : Probability, Linear Algebra, Differential Equation, Calculus
- **CS and ML Domain** : Intro to Programming(C), Foundation of Machine Learning, Data Structure and Application

PROJECTS

ChromaFormer - Hybrid Deep Learning for Epigenetic Prediction

- Developed a **hybrid BiLSTM-Transformer model** to predict gene expression from **histone modification signals** using the **Zenodo dataset 2652278.**
- Achieved an **AUC of 0.80**, performing on par with state-of-the-art models.
- Demonstrated **steadier AUC improvement per epoch**, indicating more stable convergence and robust model training

Pick My Plate - Personalised Meal Recommendation system

- Built a **deep learning-based recommendation system** to suggest personalized mess meals for IIT-H students, incorporating **user preferences, hunger levels, dietary goals, and activity context**.
- Trained models on **188 manually curated data points**, achieving **91.4% prediction accuracy** in meal recommendation tasks.
- Deployed the best-performing model using a **Firebase backend** with a **ReactJS-based web dashboard**, enabling real-time user interaction and feedback.

CRISPR-Cas9 Sequence Analysis & Off-Target Prediction

- Developed a **Python-based CRISPR-Cas9 simulation pipeline** to model guide-target interactions, cleavage probability, DNA repair outcomes, and off-target effects.
- Implemented **sequence feature extraction and probabilistic scoring**, incorporating mismatch positions, GC content, and cleavage likelihood to evaluate candidate target sites.
- Built a **Flask-based interactive application** for sequence analysis and visualization, integrating genomic FASTA inputs with automated off-target detection and risk estimation.